

Revealing the Mechanism of Sodium Diffusion in Na_xFePO_4 Using an Improved Force Field

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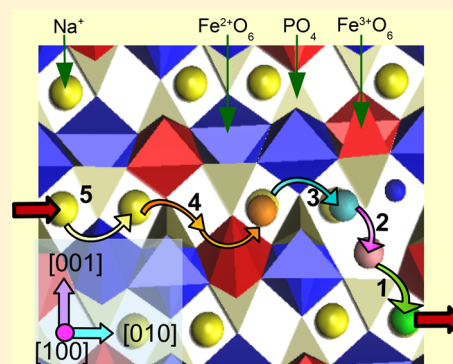
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S Supporting Information

ABSTRACT: Olivine NaFePO_4 is a promising cathode material for Na-ion batteries. Intermediate phases such as $\text{Na}_{0.66}\text{FePO}_4$ govern phase stability during intercalation-deintercalation processes, yet little is known about Na^+ diffusion in Na_xFePO_4 ($0 < x < 1$). Here we use an advanced simulation technique, Randomized Shell Mass Generalized Shadow Hybrid Monte Carlo Method (RSM-GSHMC) in combination with a specifically developed force field for describing Na_xFePO_4 over the whole range of sodium compositions, to thoroughly examine Na^+ diffusion in this material. We reveal a novel mechanism through which $\text{Na}^+/\text{Fe}^{2+}$ antisite defect formation halts transport of Na^+ in the main diffusion direction $[010]$, while simultaneously activating diffusion in the $[001]$ channels. A similar mechanism was reported for Li^+ in LiFePO_4 , suggesting that a transition from one- to two-dimensional diffusion prompted by antisite defect formation is common to olivine structures, in general.



INTRODUCTION

Today's commercial rechargeable batteries rely on Li-ion technology. Yet the expansion of the battery market toward electric vehicles and large-scale grid storage creates some important concerns.¹ One particular worry is that a massive use of lithium in the future will probably be tied to steeply increasing prices and sustainability issues. Thus, convenient alternatives to lithium are needed. From a chemical viewpoint, sodium is positioned immediately below lithium in the periodic table and, therefore, it is its natural surrogate.^{2,3} Added to this, the high abundance, environment-friendly nature, and low cost of sodium have made research in Na-based batteries a topic of high interest in recent years.^{4–12}

For the past few decades, research in Li-ion batteries has been racing to gather knowledge on the redox chemistry between lithium and a wide range of host materials (see, e.g., Nitta et al.¹³ for a brief review). Despite the chemical similarity between lithium and sodium, these accomplishments must be taken with caution when considering analogous Na-based electroactive materials. One prominent example is graphite, which is commonly used as anode in today's Li-ion batteries;^{14,15} yet graphite does not intercalate sodium under moderate conditions.^{16,17} The origin of this difference has remained a mystery for a long time. Only very recently, first-principles calculations were able to shed light into this issue by showing that Li–C bonds in graphite are enhanced due to a covalent contribution, which is missing in the case of Na–C bonds and makes Na-intercalated graphite unstable.⁸⁰ This particular example highlights two important general conclusions: (i) beyond obvious differences of mass and size between

lithium and sodium cations, the chemical behavior (i.e., electrochemical performance) of Li- and Na-intercalated compounds can also be very different due to electronic structure effects; and (ii) atomistic modeling is a useful approach to close the existing gap between our extensive knowledge on Li-based electroactive materials and equivalent Na-based ones.

Here we use atomistic modeling to examine a promising cathode material for Na-ion batteries, olivine NaFePO_4 . From a Li-based viewpoint, sustainability, and cost driven research on polyanionic iron-based materials has led us to olivine LiFePO_4 , which has clearly been the most studied one to date among commercial cathode materials.¹⁸ LiFePO_4 shows stand-out features such as high stability, high rate capability, and sustained high voltage throughout the whole discharge cycle. In principle, one could expect NaFePO_4 to inherit these properties from its isostructural lithium counterpart. Therefore, there has been a recent flurry of interest in NaFePO_4 (see, e.g., Fang et al.¹⁹). The first fundamental studies revealed that the intercalation chemistry of lithium and sodium in FePO_4 are, however, significantly different.^{20,21} These differences are generally attributed to the formation of ordered partially sodiated structures in the case of NaFePO_4 ^{19,22–24} as opposed to the single-phase behavior of LiFePO_4 .^{25–27} The formation of stable Na^+ arrangements adds therefore structural complexity to the study of sodium intercalation at the molecular level. This issue

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is particularly relevant when studying dynamic properties such as Na⁺ mobility using simulations, requiring large supercells and long simulation times to account for possible sodium orderings. These requirements make the use of first-principles methods, such as density functional theory (DFT), prohibitive in terms of computational cost.

Our study addresses this computational challenge as follows. First, we have chosen to apply an enhanced sampling technique instead of conventional molecular dynamics (MD) in order to mitigate the need for too long simulation times. In particular, we have used the Randomized Shell Mass Generalized Shadow Hybrid Monte Carlo method (RSM-GSHMC), which we developed and tested on olivine NaFePO₄ before.²⁸ Second, we have developed a robust force field for NaFePO₄ in order to effectively reduce the computational cost associated with DFT calculations when dealing with large supercells, without sacrificing accuracy and predictiveness. The proposed new force field includes polarizability effects through a core–shell model and it has specifically been designed to describe simultaneously Fe²⁺ and Fe³⁺. This is important because partially sodiated FePO₄ (which we will refer to as Na_xFePO₄) contains a mixture of Fe²⁺ and Fe³⁺ species due to the presence of Na⁺ vacancies in the structure. In this regard, the proposed force field goes beyond the previously suggested interatomic potentials for NaFePO₄, which were specifically derived to simulate fully sodiated structures and thus are only valid when one type of iron species, Fe²⁺, is present.^{29,30} Notably, our force field is parametrized using an extensive DFT-based configuration data set that covers a wide range of sodium concentrations. This aspect provides a level of predictability not available in force fields obtained from a limited amount of experimental data.

In the following, we first set out the details of the new developed force field for Na_xFePO₄. Then we thoroughly discuss accuracy and sampling efficiency aspects of our proposed simulation approach. Finally our main results are reported and analyzed on the basis of calculated Na⁺ diffusion pathways at different temperatures and sodium concentrations, comparing the performance of conventional MD and RSM-GSHMC simulations combined with the new and previously known force fields. Our findings reveal novel Na⁺ diffusion dynamics, which actively involves the formation of Na⁺/Fe²⁺ antisite defects. Such defect formation processes are generally assumed to take place during ion intercalation in polyanionic iron-based materials,^{31,32} but this is the first time that they have directly been visualized in atomistic simulations. Finally, we put forward that the accelerated sampling provided by the RSM-GSHMC method can be a very useful technique to observe, in general, this kind of rare events in solid state ionics within reasonable simulation time scales.

DEVELOPMENT OF THE NEW FORCE FIELD

Understanding Na⁺ diffusion in Na_xFePO₄ is fundamentally important when considering its use as a cathode material in Na-ion batteries. While DFT studies have provided valuable insights into a range of materials properties,^{33,34} these works are limited to short time-scales over prespecified migration paths. In order to explore longer time-scales without migration path constraints, simpler classical interatomic potentials are needed.^{35–38} The development of such potentials (i.e., a force field) has three fundamental requirements:

- an appropriate functional form of an interaction potential,
- a training data set of structural, mechanical and/or thermodynamic properties (e.g., experimental, computed, etc.) to fit the model and
- an optimization strategy to perform the fitting.

In the following subsections, these three aspects are described specifically for the development of a new force field for Na_xFePO₄.

Force Field Model. Long-range electrostatic interactions in ionic liquids or metal oxides, including all typical metal-ion cathodes, represent a challenge for the development of accurate force fields. In particular, ionic polarizability in these materials can be strong enough to significantly influence thermodynamic and transport properties.^{39–42} In this work, ionic polarizability is described through the core–shell model suggested by Dick and Overhauser,⁴³ in which a central core of a charge *X* and a shell of a charge *Y* are introduced in such a way that the sum of these charges *X* + *Y* equals to the valence state of the ion. The core and shell are coupled together in a unit via a harmonic potential, which allows the shell to move with respect to the core, thus simulating a dielectric polarization. Following Whiteside et al.,³⁰ only the Fe²⁺ and the O^{2−} ions are split into core–shell units. Hereafter, O_{sh} and Fe_{sh} will refer to the O^{2−} and Fe²⁺ shells, respectively.

The total potential energy in our model is given by

$$U_{\text{tot}} = U_C + U_{\text{Born}} + U_{\text{CS}} + U_{\text{angle}} \quad (1)$$

where *U_C* represents long-range Coulomb interactions, *U_{Born}* short-range repulsion and long-range van der Waals forces, *U_{CS}* is the interaction within core–shell units and *U_{angle}* is an angular term for the PO₄ tetrahedral units.

Coloumbic interactions are explicitly given by

$$U_C = \frac{1}{4\pi\epsilon} \sum_{i,j=1}^N \frac{q_i q_j}{r_{ij}} \quad (2)$$

where ϵ is the vacuum permittivity; *r_{ij}* is the distance between particles *i* and *j*; *q_i* and *q_j* are their respective charges; and *N* is the total number of particles. The Born interaction is described by⁴⁴

$$U_{\text{Born}} = \sum_{i,j=1}^N A_{ij} \exp\left(-\frac{r_{ij}}{\rho_{ij}}\right) - \frac{C_{ij}}{r_{ij}^6} + \frac{D_{ij}}{r_{ij}^8} \quad (3)$$

where *A_{ij}*, *ρ_{ij}*, *C_{ij}*, and *D_{ij}* are positive constants defining the shapes of the repulsive and attractive terms of the potential.

The core–shell interaction energy is estimated as

$$U_{\text{CS}} = \frac{1}{2} \sum_{l=1}^L k_l r_l^2 \quad (4)$$

where *k_l* is the spring constant for the *l*th core–shell unit, *r_l* is the core–shell distance, and *L* is the total number of shells. Importantly, neither Coloumbic nor Born interactions are computed within core–shell units. Moreover, for ions described as core–shell units only the shells are involved in the estimation of Born interactions, while the cores participate only electrostatically.

Following Whiteside et al.,³⁰ a three-body angular interaction (O_{sh}–P–O_{sh}) in the PO₄ tetrahedral units is also included

$$U_{\text{angle}} = \sum_{k=1}^K \frac{1}{2} k_{\text{ang}} (\theta_k - \theta_0)^2 \quad (5)$$

where k_{ang} is the spring constant, θ_0 is the equilibrium bond angle, θ_k is the current value of the bond k , and K is the total number of angle interactions.

Two main approaches are found in the literature to simulate core–shell systems: the so-called shell relaxation (CS-min) scheme⁴⁵ and the adiabatic shells (CS-adi) method.³⁹

In the CS-min scheme an energy minimization is performed after each MD step in order to update the shell positions, making this approach very computationally demanding. Conversely, in the CS-adi method, a small fraction y of the ion mass is put on the shell, whereas the remaining $1 - y$ fraction is left to the core. Then, all the particle positions propagate following the conventional MD technique. For a sufficiently small value of y , the shells adiabatically follow the cores motion during the simulation. However, the period of the core–shell spring is proportional to $\sqrt{y(y-1)}$,³⁹ which means that a small y requires a small integration time-step. In recent work from this group, a systematic methodology was proposed to introduce a shell mass in a way that minimizes the negative effect on the kinetic energy of the system.²⁸ This methodology is used in the present study.

Parameter Estimation with *potfit*. The program *potfit*^{46,47} was chosen to parametrize the proposed force field. *potfit* makes use of the force-matching method,⁴⁸ in which the potential parameters are adjusted to optimally reproduce forces, stresses and energies from reference data obtained through first-principles calculations.

All reference configurations in this study were computed using the plane wave DFT code VASP.^{49,50} For configuration m consisting of N_m particles, VASP calculates one energy e_m^0 , six components of the stress tensor $s_{m,l}^0$ ($l = 1, \dots, 6$) and $3N_m$ Cartesian force components, $f_{m,n}$ ($n = 1, \dots, 3N_m$). The details of the generated DFT database are provided in the next subsection, while the cost function implemented in *potfit*^{46,47} is described in section S1 of the SI.

Instead of allowing all parameters to freely vary, only the van der Waals parameters between oxygen shells ($\text{O}_{\text{sh}}\text{--}\text{O}_{\text{sh}}$) and those from the angular contribution ($\text{O}_{\text{sh}}\text{--}\text{P}\text{--}\text{O}_{\text{sh}}$) were estimated. All other parameters were left as originally proposed by Whiteside et al.³⁰ It was not necessary to modify the parameters for other species because the most significant effect observed when removing Na^+ from the NaFePO_4 system was an anomalous displacement of the $(\text{PO}_4)^{3-}$ units (see section S3 in the Supporting Information (SI)). Hence, training the Born potential in order to capture the $\text{O}_{\text{sh}}\text{--}\text{O}_{\text{sh}}$ repulsion in the presence of a neighboring vacancy was enough for producing a robust force field applicable to materials with different sodium concentrations.

It is important to properly select the initial location of the shells prior to the minimization procedure. In section S2 of the SI a simple and consistent solution to perform this task is provided. Finally, we must note that the *potfit* package was modified to introduce the Born model, the damped shifted force method for the Coulomb sum and the three-body angular interactions. In addition, we tuned some built-in parameters as described in the SI and introduced the option to ignore Coulomb interactions within the core–shell units.

Development of the DFT Database. We generated a set of 58 olivine Na_xFePO_4 configurations consisting of (a) 11

geometrically optimized structures at 0 K, ranging from 72 to 84 atoms (3 unit cells in the [010] direction) and (b) 58 equilibrated snapshots taken out of ab initio MD trajectories, from an ideal NaFePO_4 crystal at 500 and 1000 K.

To obtain the geometrically optimized structures, spin-polarized DFT calculations were performed as implemented in the VASP code^{50,51} at different sodium concentrations: $x = 1.0, 0.83, 0.66, 0.58$, and 0.0 . We used as a reference the published convex hull of Na_xFePO_4 at 0 K.⁵² As shown by Saracibar et al.,⁵² the intermediate phases at $x = 0.83$ and $x = 0.66$ govern phase stability during intercalation/deintercalation. Moreover, for $x > 0.5$ certain structures display negative formation energies and thus are likely to exist in a metastable form for brief periods of time. On the other hand, all structures with $x < 0.5$ are unstable. Many distinct supercells can be built satisfying $x = 0.83, 0.66$, and 0.58 . Hence, for every x we chose the three configurations having the lowest formation energies as reported by Saracibar et al.⁵²

Projector augmented wave potentials⁵³ were employed to replace the inner electrons, whereas Na (3s), Fe (3p, 3d, 4s), P (3s, 3p), and O (2s, 2p) valence electrons were expanded in plane-waves with a cutoff energy of 600 eV. The Perdew–Burke–Ernzerhof (PBE)⁵³ exchange–correlation functional was used together with a Monkhorst–Pack grid with at least $2 \times 4 \times 5$ k-point sampling per $1 \times 1 \times 1$ unit cell. In order to describe the localized Fe 3d states a Hubbard U-like term was added to the PBE functional.⁵⁴ Here we used a value of $U = 4.3$ eV for Fe atoms, as proposed by Zhou et al.⁵⁵ for LiFePO_4 . The 11 structures were fully optimized (cell parameters, volume cells, and atomic positions) with a residual force threshold of 0.02 eV/Å. These computational settings guarantee a tight convergence in total energies (less than 5 meV per formula unit).

Starting from the optimal olivine NaFePO_4 , ab initio MD simulations were performed at 500 and 1000 K using VASP.^{50,51} The model system was first thermally equilibrated in a microcanonical ensemble (NVE) for 2 ps and then run for 20 ps in a canonical ensemble (NVT) at the chosen temperatures controlled by a Nose Hoover thermostat.⁵⁶ 42 configurations at 500 K and 16 configurations at 1000 K were extracted from the production runs and subsequently incorporated into the training data set.

Optimization of the force field parameters with *potfit* completed with the following root-mean-square errors (eq S1 in the SI): $\Delta F_f = 0.32$ eV/Å, $\Delta F_s = 0.014$ eV/Å³, and $\Delta F_e = 1.49$ eV. The resulting parameters are listed in Table 1. We refer to the force field developed here as $\text{Na}_x\text{FePO}_4\text{-ff}$, whereas that from Whiteside et al.³⁰ as Whiteside-ff. It is interesting to note that the van der Waals parameters for $\text{Fe}^{2+}\text{--}\text{O}_{\text{sh}}^{2-}$ worked very well to describe the $\text{Fe}^{3+}\text{--}\text{O}_{\text{sh}}^{2-}$ interaction. It is possible that optimizing the $\text{Fe}^{3+}\text{--}\text{O}_{\text{sh}}^{2+}$ parameters could lead to some slight improvement and, indeed, we tried this alternative. However, the fact that the concentrations of Fe^{3+} depends on x makes such optimization particularly complex. In our case, given that more configurations are available at high x in the data set, the parameters for Fe^{3+} obtained through simulated annealing favored richly sodiated configurations. On the other hand, those with low x expanded unrealistically due to excessive $\text{Fe}^{3+}\text{--}\text{O}_{\text{sh}}^{2-}$ van der Waals repulsion. Focusing only on $\text{O}_{\text{sh}}^{2-}\text{--}\text{O}_{\text{sh}}^{2-}$ and $\text{O}_{\text{sh}}^{2-}\text{--}\text{P}\text{--}\text{O}_{\text{sh}}^{2-}$ (P and $\text{O}_{\text{sh}}^{2-}$ concentrations are independent of x) leads to excellent results without the need for a much larger database.

Table 1. Force Field Parameters for Olivine Na_xFePO_4 Obtained with *potfit* ($\text{Na}_x\text{FePO}_4\text{-ff}$)^a

Born–Mayer–Huggins				
interactions	A (eV)	ρ (Å)	C (eV Å ⁶)	D (eV Å ⁸)
Na ⁺ –O _{sh} ^{2–}	629.768	0.317	0.0	0.0
Fe _{sh} ²⁺ –O _{sh} ^{2–}	1105.241	0.311	0.0	0.0
Fe _{sh} ³⁺ –O _{sh} ^{2–}	1105.241	0.311	0.0	0.0
P ⁵⁺ –O _{sh} ^{2–}	897.265	0.358	0.0	0.0
O _{sh} ^{2–} –O _{sh} ^{2–}	1506.355	0.100	36.212	1.01
	(22764.3)	(0.149)	(44.53)	(0.0)
Core–Shell				
species	core charge	shell charge	k (eV Å ^{–2})	
Fe ²⁺	–0.997	2.997	19.26	
O ^{2–}	0.96	–2.96	65.0	
Angular				
bond	k_{ang} (eV rad ^{–2})		θ_0 (deg)	
O _{sh} ^{2–} –P ⁵⁺ –O _{sh} ^{2–}	4.159		109.47	
	(1.323)		(109.47)	

^aOnly the parameters corresponding to the $\text{O}_{\text{sh}}-\text{O}_{\text{sh}}$ and $\text{O}_{\text{sh}}-\text{P}-\text{O}_{\text{sh}}$ differ from those estimated by Whiteside et al., which are reported in parentheses.

In the coming sections, we proceed to perform atomistic simulations using different techniques, such as MD and RSM-GSHMC, to validate and test the new force field. The validation includes comparison with DFT simulations, other force fields and experimental results when available.

■ VALIDATION

We check accuracy and performance of the newly proposed force field by comparing the computed structural parameters, unit cell volumes and diffusion coefficients against those obtained with Whiteside-ff, DFT, and experiments when varying temperature and Na^+ concentration.

Methodology and Simulation Setup. Both force fields presented in Table 1 were implemented in the open source MD software MultiHMC-GROMACS. This is an in-house software package for atomistic simulations based on GROMACS version 4.5.4,⁵⁷ which was modified to use hybrid Monte Carlo based methods^{58–60} and multi-stage integration schemes.^{61–63} It is available for public use under the GNU Lesser General Public License.

As samplers, we alternatively used standard MD with a velocity Verlet integrator and RSM-Generalized Shadow Hybrid Monte Carlo (RSM-GSHMC)²⁸ with the two-stage MAIA integrator.⁶³

The GSHMC method is a type of hybrid Monte Carlo (HMC) which offers several improvements over standard HMC and MD by using modified energies instead of true Hamiltonians for sampling, in order to increase the sampling efficiency. The modified energies or shadow Hamiltonians are better conserved by symplectic integrators, leading to a faster convergence of GSHMC compared with HMC. The partial momentum update introduced in GSHMC allows for retaining dynamical information, and thus makes the method applicable to simulation of diffusion processes. The GSHMC method is particularly useful when studying multi-dimensional spaces and rare events. In this study it is chosen to improve the sampling of the ionic diffusivity of Na_xFePO_4 . The details of the GSHMC method and its implementation can be found elsewhere.^{60,64,65,70–72}

In RSM-GSHMC, a mass randomization is applied to the masses of the shells $\text{O}_{\text{sh}}^{2-}$ and $\text{Fe}_{\text{sh}}^{2+}$. This helps to reduce any potential negative effects on the kinetic properties of the system when using an adiabatic core–shell model. In this study, RSM-GSHMC is combined with the MAIA integrator which provides the highest accuracy among all other two-stage splitting methods including Verlet.

Prior to performing the simulations, all systems were first stabilized using the steepest descent method for energy minimization. This was followed by 50 ps of equilibration at constant temperature and another 50 ps at constant pressure. The Andersen barostat⁶⁶ was used to impose constant pressure for both MD and GSHMC, as implemented in MultiHMC-GROMACS.⁶⁵ In MD we used a Nose-Hoover thermostat,⁶⁶ while the GSHMC method keeps temperature by design.⁶⁰ We used the force field parameters presented in Table 1 with a cutoff of 12 Å for electrostatics. Periodic boundary conditions were applied in the three dimensions. The production simulations for various sodium concentrations between $x = 0$ and 1 were run in the NPT ensemble over 10 ns at temperatures ranging between 10 and 900 K, at a pressure of 1 bar using a time step of 1 fs. For all the results presented in this section, we employed 4th order shadow Hamiltonians and fixed the GSHMC parameters to $L = 500$ steps for trajectory lengths between Monte Carlo steps and to $\phi = 0.1$ for the partial momentum update.

Accuracy. In order to assess the accuracy of the newly proposed force field, we measure the unit cell parameters for Na_xFePO_4 after a 3 ns simulation using RSM-GSHMC at a constant pressure of 1 bar and a temperature of 300 K. Simulations are performed for systems with Na^+ concentrations $x = 0, 0.58, 0.66, 0.83$, and 1. In Table 2 the simulation results

Table 2. Computed Lattice Constants of Olivine Na_xFePO_4 Using RSM-GSHMC for Varying Concentrations of Na^+ at 300 K with Both Force Fields^a

	param. (Å)	exp	DFT	Whiteside-ff	$\text{Na}_x\text{FePO}_4\text{-ff}$
FePO_4	a	9.81	9.97		9.49
	b	5.79	5.92		5.78
	c	4.78	4.87		4.74
$\text{Na}_{0.66}\text{FePO}_4$	a	10.28	10.41	10.13	10.33
	b	6.08	6.06	6.08	6.03
	c	4.93	4.99	5.09	5.01
$\text{Na}_{0.83}\text{FePO}_4$	a	10.31	10.45	10.49	10.37
	b	6.11	6.19	6.08	6.23
	c	4.95	4.99	5.03	4.94
NaFePO_4	a	10.41	10.50	10.32	10.45
	b	6.22	6.27	6.33	6.26
	c	4.95	4.99	5.07	4.99

^aExperimental and DFT values are taken from Saracibar et al.⁵²

alongside with the experimental and DFT values⁵² are shown. Data for $x = 0.58$ are not included given the lack of experiments and independent DFT values to compare with. Nevertheless, it is worth mentioning that simulations using Whiteside-ff crash at this concentration.

We found that the divergence between the unit cell parameters calculated using two different force fields was small, although $\text{Na}_x\text{FePO}_4\text{-ff}$ achieved values slightly closer to the experimental results. It was also found that $\text{Na}_x\text{FePO}_4\text{-ff}$ succeeded in simulations of a wider variety of sodium ion concentrations, while simulations using Whiteside-ff became

numerically unstable for $x < 0.66$ (see Figure S3 in SI). Instability in the Whiteside-ff is not surprising since the parameters were obtained by fitting with respect to the experimental bond lengths and unit cell parameters of fully sodiated NaFePO_4 . In this regard, there is no expectation for this force field to be applicable at lower levels of sodiation. In particular, the presence of sodium vacancies are likely to produce instabilities in systems simulated with Whiteside-ff, given that this force field does not account for the presence of Fe^{3+} .

In Figure 1 the volume expansion for a unit cell is presented for temperatures from 10 to 900 K for a fully sodiated

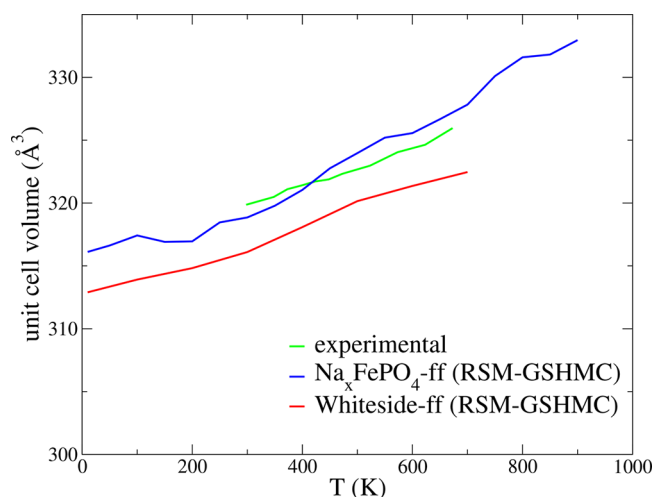


Figure 1. Unit cell volume expansion for temperatures from 10 to 900 K for a fully sodiated NaFePO_4 system computed using RSM-GSHMC with Whiteside-ff and Na_xFePO_4 -ff. Experimental results are taken from Moreau et al.⁶⁷ For temperatures above 700 K, Whiteside-ff could not produce numerically stable simulations.

NaFePO_4 system computed with RSM-GSHMC using both Whiteside-ff and Na_xFePO_4 -ff. One can see that the two force fields produce similar trends, although the unit cell volume calculated with Na_xFePO_4 -ff is generally higher and in better agreement with the experimental results taken from Moreau et al.⁶⁷ We notice that Whiteside-ff provides a slightly more accurate estimate of dV/dT than Na_xFePO_4 -ff, which is of interest for computing the thermal expansion coefficient. However, for the calculation of the diffusivity, we expect Na_xFePO_4 -ff to perform better, because at this level of confinement, diffusion is extremely sensitive to small changes in the cross section area of the transport channels and the sites volume (see, e.g., Krishna and Van Baten⁶⁸). It is also important to mention that Whiteside-ff at temperatures above 700 K could not produce numerically stable simulations as a result of exceedingly large forces between oxygen shells.

Sampling Efficiency. Once it has been established that Na_xFePO_4 -ff can reproduce structural parameters accurately, we measure its performance by studying the rare events dynamics of Na^+ diffusion in Na_xFePO_4 . For the following simulations, we focus on $x = 0.66$, as this ratio between ions and vacancies is a stable intermediate phase and, therefore, interesting for the study of charge transport dynamics during cycling.⁵² The diffusion coefficient is a notoriously difficult property to measure in solid state atomistic simulations, requiring exceedingly large systems and very long simulated times. Thus, a force field which can more efficiently replicate the rare jumps of Na^+

ions along the crystallographic structure combined with an enhanced sampling algorithm would be highly advantageous.

Simulations of 10 ns were performed with temperatures ranging from 10 to 900 K using RSM-GSHMC combined with the two force fields and MD with Na_xFePO_4 -ff. Diffusion at temperatures below 300 K was very difficult to observe, indicating that even longer simulations might be required. However, it was found that the number of ionic jumps is higher when using Na_xFePO_4 -ff instead of Whiteside-ff at temperatures above 550 K. For instance, at 600 K, each Na^+ performs in average ~ 4 jumps/ns using Na_xFePO_4 -ff, while only ~ 2 jumps/ns using Whiteside-ff. Na_xFePO_4 -ff does not lead to numerical instabilities, although at temperatures above 900 K, the system becomes amorphous. This is in agreement with experimental observations using X-ray diffraction.⁶⁷

The diffusion coefficients are estimated from the mean square displacements (MSD) according to⁵⁷

$$D = \lim_{t \rightarrow \infty} \frac{1}{2nt} \langle \|\vec{r}_i(t) - \vec{r}_i(0)\|^2 \rangle_{i \in \{\text{Na}^+\}} \quad (6)$$

where n is dimensionality (1, 2, or 3 for one, two, or three-dimensional diffusion, respectively), $\vec{r}_i$ is the position of atom i and $\{\text{Na}^+\}$ is the set of sodium atoms in the simulation box.

Figure 2 shows the three-dimensional diffusion coefficients of Na^+ for a range of temperatures using RSM-GSHMC with both

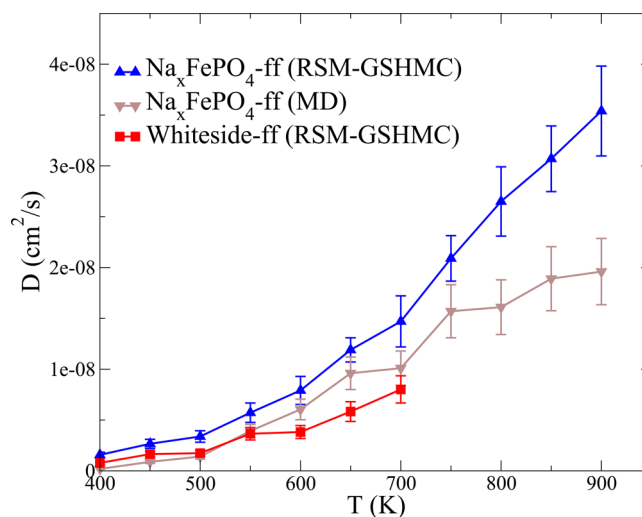


Figure 2. Three-dimensional diffusion coefficient and corresponding error bars at temperatures between 200 and 900 K for a $\text{Na}_{0.66}\text{FePO}_4$ system using both Whiteside-ff and Na_xFePO_4 -ff. The divergence observed at high temperatures is a consequence of the improved sampling efficiency of the chosen sampler. At temperatures above 700 K, Whiteside-ff could not produce numerically stable simulations.

force fields. We also plot results for similar simulations using standard MD with Na_xFePO_4 -ff. Comparison of the three curves suggests that not only does the force field affects the diffusion coefficients but the sampling efficiency of the chosen sampler also does. Indeed, at higher temperatures the computed diffusions using MD and RSM-GSHMC combined with the same force field (for equal simulation times) diverge visibly. This is a consequence of the higher sampling efficiency of GSHMC methods.⁶⁰ For longer simulation times, both methods are expected to converge.²⁸

This is also confirmed by Figure 3, which presents the MSD for Na^+ at 700 K in the main diffusion direction, [010],

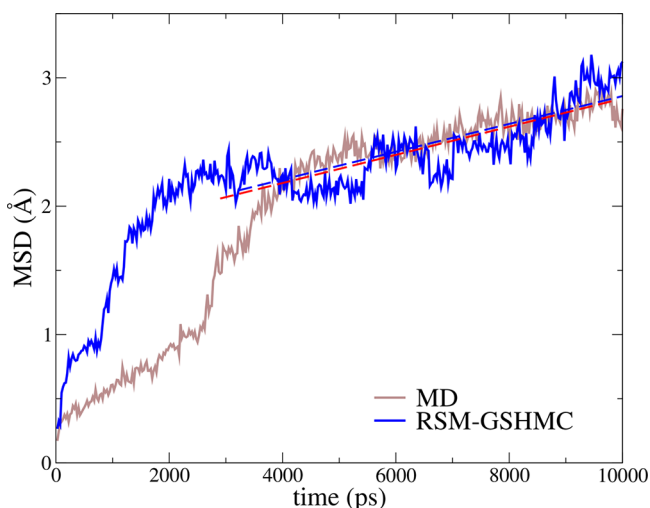


Figure 3. MSD at 700 K in the main diffusion direction, [010], using both traditional MD and RSM-GSHMC combined with Na_xFePO_4 -ff. Convergence to the equilibrium slope is reached faster through RSM-GSHMC.

obtained from both traditional MD and RSM-GSHMC. Convergence to the equilibrium slope that provides the diffusion coefficient is reached nearly 2 ns quicker through RSM-GSHMC, demonstrating its ability to sample the configurational space more efficiently than MD. We must stress that Figure 2 depicts the three-dimensional diffusion coefficient, while Figure 3 depicts specifically the diffusivity along the [010] direction. Therefore, they are not necessarily equal. In particular, transport along the [100] axis is slightly more favored when using RSM-GSHMC, and this is reflected in Figure 3.

The noise in the MSD curves in Figure 3 is at least partially due to the fact that a fraction of the Na^+ atoms do not perform any jump or jump only once, as we discuss later. While the situation could be improved by extending the simulations, the fact that the polarizable core-shell model involves a very light particle (the shell), severely limits the simulation time-step, and further increasing the simulation time comes with a significant increase in the computation time.

RESULTS AND DISCUSSION

Study of Diffusion of Na^+ in $\text{Na}_{0.66}\text{FePO}_4$. Since Na^+ diffusion is an activated process with an energy barrier of 0.32–0.44 eV,^{29,33,34} its study cannot be easily accessed through classical MD methods. Moreover, the absence of suitable force fields for $x < 1$ has constrained previous studies to fully sodiated NaFePO_4 by creating few Na^+ vacancies in the lattice.²⁹ First-principles analysis of diffusion in $\text{Na}_{0.93}\text{FePO}_4$ have determined that Na^+ transport is limited to the [010] direction on a curved path that is perpendicular to the faces below the shared edges of NaO_6 tetrahedra (Figure 4). The estimated activation energies range from 0.38 to 0.44 eV.^{33,34} A lower activation energy of 0.32 eV was obtained by Tripathi et al.²⁹ using the Mott-Littleton scheme incorporated in the GULP code⁶⁹ and Whiteside-ff.

We use the RSM-GSHMC hybrid Monte Carlo scheme^{28,60} to explore for the first time diffusion of Na^+ in $\text{Na}_{0.66}\text{FePO}_4$, a relevant intermediate phase during the intercalation/deintercalation process. The previous study using Whiteside-ff has demonstrated the ability of RSM-GSHMC to improve sampling

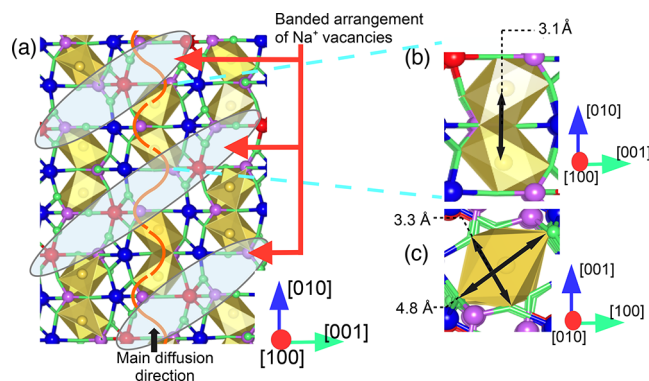


Figure 4. (a) Structure of olivine $\text{Na}_{0.66}\text{FePO}_4$. The octahedral coordination around Na^+ are shown in yellow, while the remaining atomic species are color coded as follows: O^{2-} in green, P^{5+} in purple, Fe^{2+} in blue, and Fe^{3+} in red. The curved trajectory followed by $\text{Na}^{+30,34}$ along one of the channels in the main diffusion direction, [010], is depicted as the solid orange line. Na^+ vacancies and Fe^{3+} ions follow the banded arrangement displayed in the figure. (b) Linear distance between Na^+ sites in the [010] direction is 3.1 Å. This is the length of individual Na^+ jumps. (c) Cross section of a diffusion channel along [010]. The dimensions of the channel are measured between oxygen centers. The channels along [001] are much narrower, suggesting that jumps in this direction would be rare in defect free structures. The [100] is fully blocked.

over traditional MD in NaFePO_4 .²⁸ Hence, the method's efficiency has been validated for this type of systems and the relatively small changes in the force field proposed here are not expected to deteriorate it.

Using classical MD, Boulfelfel et al.⁷³ showed that, at 1200 K, it takes an average of 1200 ps to generate a single Frenkel defect in LiFePO_4 . A similar time scale is reasonably expected to operate for NaFePO_4 : while Na^+ ions are larger than Li ions, the channel cross section in NaFePO_4 is larger than that in LiFePO_4 , somewhat compensating the ion size effect.²⁹ Consequently, long 10 ns simulations were performed at 300, 500, and 700 K.

Figure 5a depicts the maximum displacement per particle (MDPP) during a 10 ns run using RSM-GSHMC with Na_xFePO_4 -ff and Whiteside-ff. The dashed horizontal lines represent displacements a particle must perform in order to jump to an adjacent cavity. Notably, in comparison to Whiteside-ff, Na_xFePO_4 -ff allows for a larger proportion of particles (~20%) to jump between cavities over the simulation time, thus producing a higher diffusion coefficient (Figure 5b,c). This does not necessarily mean that Whiteside-ff produces less accurate diffusivity results. However, the underestimated volumetric expansion of the structure, observed in Figure 1 at 700 K with Whiteside-ff, may, indeed, lead to slightly narrower channels and more constrained displacements and, as a result, to the underestimated diffusivity.

Figure 6a shows the MDPP after a 10 ns RSM-GSHMC run with Na_xFePO_4 -ff at 300 and 500 K. As expected, the number of hopping particles is significantly lower compared to that at 700 K (see Figure 5). Nevertheless, the MSD in the [010] direction presented in Figure 6b is monotonically linear and provides a measure of the diffusion coefficient. Interestingly, the slope of the MSD curve in the [001] direction is considerably lower than that at 700 K, suggesting that a higher temperature promotes transport through the narrower paths perpendicular to the main diffusion axis. This behavior was also captured in MD simulations of LiFePO_4 at 500 K,⁷³ occurring at a rate of 1

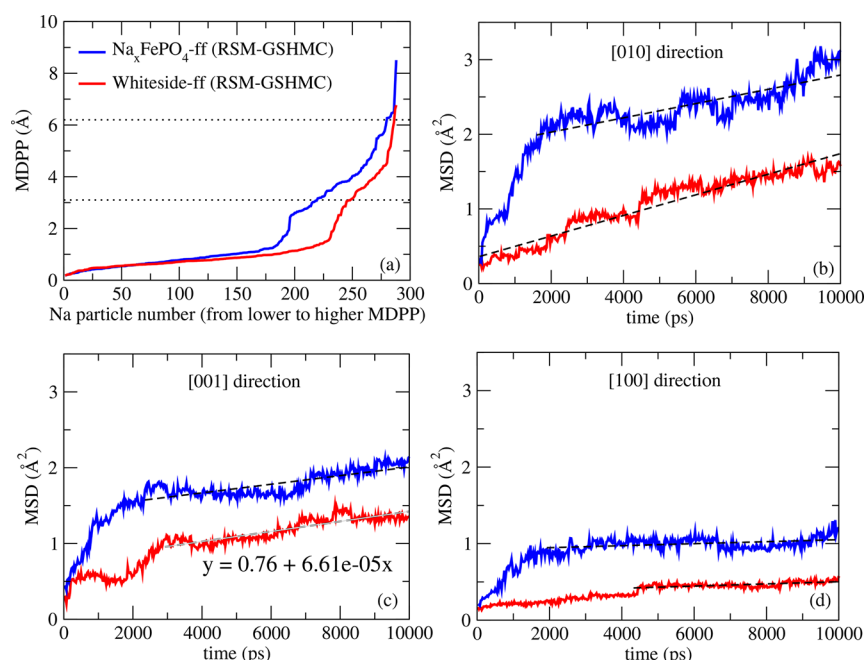


Figure 5. 10 ns simulation of Na_{0.66}FePO₄ at 700 K in the NPT ensemble using RSM-GSHMC with Na_xFePO₄-ff (blue) and Whiteside et al.-ff (red). (a) MDPP during the 10 ns run. The particles are ordered from minimum to maximum displacement. The dashed lines represent displacements that must involve jumps between Na⁺ sites. (b), (c), and (d) show the MSD vs time along the [010], [001], and [100] directions, respectively.

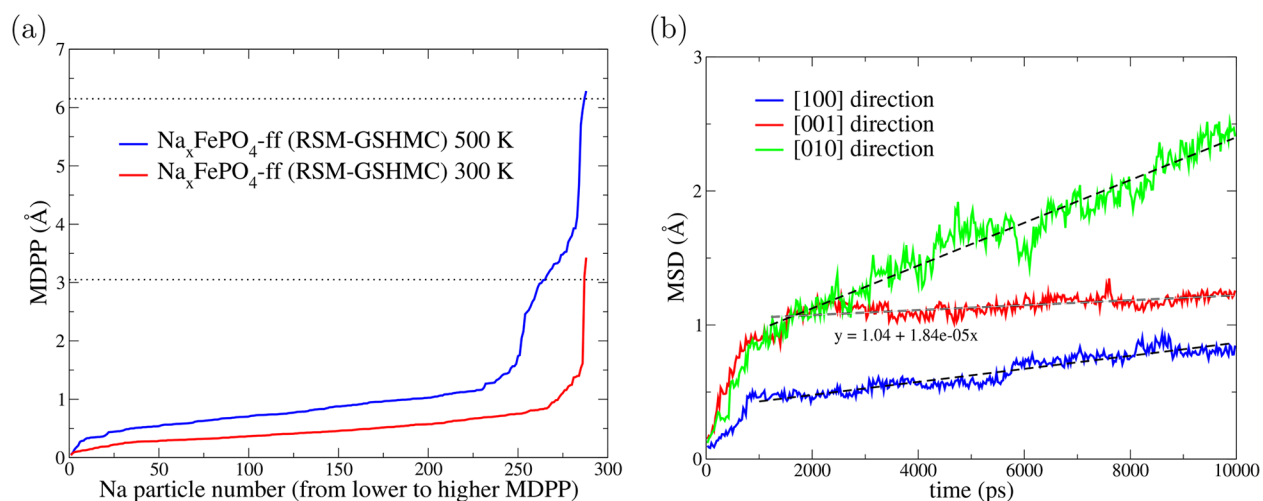


Figure 6. (a) MDPP after a 10 ns run at 300 and 500 K. (b) MSD vs time along the [010], [100], and [001] directions for 10 ns simulation at 500 K.

or 2 events every 60 ps. In our case, this is potentially a consequence of the volumetric expansion of the structure together with thermally induced disorder. As a matter of fact, the heat treatment is known to transform olivine NaFePO₄ into the more thermodynamically stable maricite phase.⁷⁴

At 300 K, diffusion is severely limited. Galvanostatic measurements estimate a diffusion coefficient of 10^{-17} cm²/s,²⁰ substantially below what can be probed through MD simulations. While this macroscopic result likely incorporates a number of additional resistances, it is evident that considerably larger simulation times are required to obtain an accurate estimate.

MD Trajectories Analysis at 700 K. During an MD run, it is important to distinguish between two types of vacancies in the structure of olivine Na_xFePO₄. On the one hand, there are *extrinsic vacancies* associated with the absence of Na⁺ with

respect to the stoichiometric structure. These vacancies are located in bands along the [011] direction and separated by Fe³⁺O₆ octahedra, as shown in Figure 4. On the other hand, there are *intrinsic vacancies* associated with the formation of Frenkel defects during Na⁺ diffusion.

Figure 7 displays an interesting trajectory observed at 700 K. It also presents the typical passage of Na⁺ through the extrinsic vacancies, not previously observed in MD studies. In the past, Tealdi et al.⁷⁵ simulated diffusion in Li_{0.75}FePO₄ using classical MD by assigning a formal charge of +2.25 to all Fe ions. Because of the uniformity in the Fe charges, the initial Li vacancies were distributed randomly, and their positions could vary freely over the course of the simulations. Our approach is fundamentally different as it distinguishes between Fe²⁺ and Fe³⁺, putting stringent constraints on the possible location of the extrinsic vacancies (see Figure 4). Over the course of our

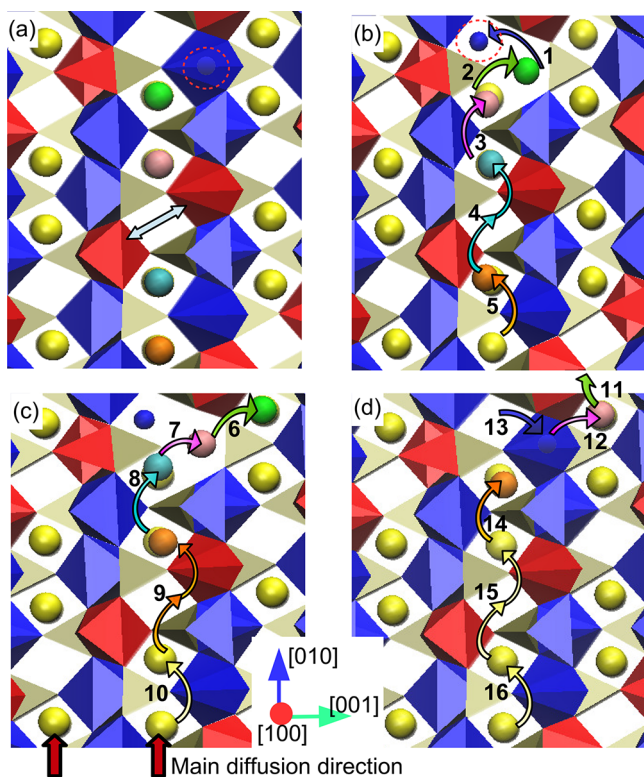


Figure 7. Two-dimensional trajectory at 700 K. At this temperature, the coordination polyhedra are slightly distorted. The red octahedra are Fe^{3+}O_6 units, the blue octahedra are Fe^{2+}O_6 units, and the tetrahedra correspond to $(\text{PO}_4)^{3-}$ units. The green, pink, cyan and orange atoms are tagged Na^+ ions; the remaining Na^+ ions are colored in yellow. The numbers in the figure specify the order in which atomic jumps occurred. (a) Initial, defect-free configuration. The double arrow shows the diametrically opposed Fe^{3+} ions around an extrinsic vacancy. (b) Antisite-like defect. A Fe^{2+} atom (encircled in red in (a) and (b)) moves to the adjacent extrinsic vacancy (jump 1), allowing the green Na^+ to jump to the empty Fe^{2+} site (jump 2). (c) Since the central diffusion channel is blocked, the green Na^+ transfers to an adjacent channel (jump 6), leading to the cascade of jumps 7–10. (d) Pink and green Na^+ continue diffusing through the adjacent channel. After the defect in the Fe^{2+} site is removed (jump 13), one-dimensional diffusion is reestablished. Notice that Na^+ diffuses through the extrinsic vacancies in single, long curvilinear jumps (4, 9, and 15).

simulations extrinsic vacancies remain fixed, perturbed only by the temporary presence of Na^+ and Fe^{2+} ions. Hence, Tealdi et al. study represents the limit of very fast polaron diffusion, while our study is concerned with the limit of very slow (with respect to Na^+ mobility) polaron diffusion.

The presence of four diametrically opposed Fe^{3+} ions around extrinsic vacancies, as indicated by the double arrow in Figure 7a, makes this type of vacancies highly repulsive to Na^+ . Even under these circumstances, formation of an intrinsic vacancy due to diffusion of the pink Na^+ (jump 3 in Figure 7b) immediately prompts the rapid transport of the cyan-labeled particle through the extrinsic vacancy, following the S-shaped trajectory presented in jump 4 of Figure 7b. Similarly, diffusion of the cyan Na^+ to the adjacent site, as depicted in jump 8 of Figure 7c, activates the rapid transport of the orange ion through the extrinsic vacancy, following the S-shaped trajectory shown in jump 9 of Figure 7c. In both cases, the residence time of Na^+ in the extrinsic vacancy site is below 2 ps, which means that the

entire movement, ~ 6.2 Å in length, can essentially be considered as a single long jump.

$\text{Na}^+/\text{Fe}^{2+}$ “antisite” pair defects occur when one Na^+ (ionic radius 1.02 Å) and one Fe^{2+} (ionic radius 0.78 Å) are interchanged between their two nonequivalent M1 and M2 octahedral sites. For olivine LiFePO_4 , the prevalence of $\text{Li}^+/\text{Fe}^{2+}$ antisite defects is ~ 1 –2% at room temperature.^{31,32,76,77} Since there is a considerable difference in radius between $\text{Na}^+/\text{Fe}^{2+}$ compared to $\text{Li}^+/\text{Fe}^{2+}$ (ionic radius 0.74 Å for Li^+), it is expected for $\text{Na}^+/\text{Fe}^{2+}$ antisites to be far less frequent. However, classical atomistic simulations indicate that the $\text{Na}^+/\text{Fe}^{2+}$ antisite formation energy (0.86 eV) is comparable to that of $\text{Li}^+/\text{Fe}^{2+}$ antisite (0.74 eV).²⁹ Hence, the prevalence of $\text{Na}^+/\text{Fe}^{2+}$ antisites in NaFePO_4 is also likely to be ~ 1 –2%. Notice that the presence of these defects blocks Na^+ diffusion along the [010] direction.

Let us refer again to LiFePO_4 . For this cathode, computational and experimental studies have shown that nanosize particles (100 nm) would be delithiated in less than 0.01 s, given that the [010] channels are nearly defect-free. Even with only one defect per channel, there would still be accessible charge carriers from the particle surface. In contrast, conductivity studies performed by Amin et al.⁷⁸ on macroscopic (millimeter scale) LiFePO_4 single crystals produce much lower diffusivities and more isotropic transport (i.e., diffusion perpendicular to the [010] channel becomes more likely). DFT studies covering an exhaustive number of diffusion paths⁷⁹ produced 2D diffusivities (along the [010] and [001] directions: y and z axes in Figure 7) that approximate better those experimentally observed in macroscopic LiFePO_4 than DFT-based diffusivities along [010] only.

While the effect of anisotropic diffusion in NaFePO_4 has not been as extensively studied, the large discrepancies between the theoretically estimated one-dimensional diffusion coefficient ($\sim 10^{-10}$ cm²/s³⁴) and that obtained from macroscopic measurements ($\sim 10^{-17}$ cm²/s²⁰) indicate that, on large scale devices, it could be quite significant. Our simulations show, for the first time, that the blockage of the main diffusion channel ([010] direction) by Fe^{2+} can in fact estimate mobility in the [001] direction, potentially reducing the overall two-dimensional diffusivity but allowing sodiation/desodiation to continue despite the temporary or permanent Fe^{2+} blockage of a diffusion channel. Figure 7 illustrates the mechanics of this process. A Fe^{2+} ion can temporarily occupy an extrinsic vacancy (jump 1, Figure 7b), leading to the formation of an antisite-like defect when the nearest Na^+ ion (green particle) moves to the free Fe^{2+} site (jump 2 in Figure 7b). Given the difference in size between Na^+ and Fe^{2+} , it is likely that the local structural deformation permits the transfer of the green Na^+ to the adjacent channel (jump 6 in Figure 7c) while simultaneously allowing the blocking Fe^{2+} to remain for several tens of ps in the highly repulsive extrinsic vacancy. A second sodium transfers to the adjacent channel (pink particle, jump 12 in Figure 7d) before the Fe^{2+} is reabsorbed into its original site (jump 13 in Figure 7d). Upon reabsorption, diffusion continues in the preferential direction.

This result is in general agreement with simulations of two-dimensional diffusion in LiFePO_4 .^{73,75,79} occasional jumps of charge carriers in the [001] can overcome (at a cost of reduced diffusivity) the damaging effect of antisite defects. Notably, we did not observe this phenomenon at 300 K. Nonetheless, this could be the result of insufficient simulation times. All other diffusion events follow the trajectories extensively described in

the literature.^{29,33,73} We believe that increasing simulation times along with the novel simulation techniques such as RSM-GSHMC coupled with Na_xFePO_4 -ff for the analysis of rare events will allow unveiling the interesting and yet vastly understudied phenomenon of the filling of extrinsic vacancies during actual intercalation/deintercalation processes.

CONCLUSIONS

In this work we developed and validated a polarizable force field for olivine Na_xFePO_4 . The new force field outperforms existing alternatives in terms of thermal stability, accuracy, and applicability to the whole range of sodium compositions ($0 < x < 1$). In addition, we applied an enhanced sampling methodology based on a shadow hybrid Monte Carlo technique, RSM-GSHMC,²⁸ in order to access the dynamics of rare events using shorter simulations and lower temperatures than those typically required by atomistic simulations on similar systems. The combination of the new force field and the RSM-GSHMC approach allowed us to study with unprecedented detail Na-ion diffusion in $\text{Na}_{0.66}\text{FePO}_4$ at the molecular-level. We confirmed that the main diffusion mechanism involves single Na-ion hops through the one-dimensional channels along the [010] crystallographic direction. Furthermore, we identified the novel Na-ion diffusion dynamics involving the formation and annihilation of $\text{Na}^+/\text{Fe}^{2+}$ antisite defects, which effectively facilitate the migration of Na-ions between adjacent [010] channels. $\text{Na}^+/\text{Fe}^{2+}$ antisite defects were expected to block [010] channels and, therefore, hinder Na-ion mobility in the bulk material. In contrast, our results revealed that such defects can indeed favor Na-ion exchange between parallel [010] channels. In principle, similar ion migration mechanisms could operate in other olivine framework compounds as well. Overall, this study provides an in-depth understanding of Na-ion mobility in olivine Na_xFePO_4 , a promising alternative to the commercially available LiFePO_4 cathode material for Na-ion batteries, and paves the way to unveil fundamental aspects of ion dynamics in polyanionic materials, in general.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b00230.

Tuning parameters for the simulated annealing algorithm incorporated in *potfit*, as well as the fitting errors are listed in section S1. A new methodology for selecting the position of the shells in the core-shell model is described in section S3. A comparison between $g_{p-p}(r)$ in the $\text{Na}_{0.58}\text{FePO}_4$ system obtained with Whiteside-ff and Na_xFePO_4 -ff is shown in section S3. Finally, the relative error in cell volumes obtained with Whiteside-ff and Na_xFePO_4 -ff is shown in section S4 (PDF).

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Notes

The authors declare no competing financial interest.

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